

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 77. Case Conversion Methods

Section 1: Lecture Summary

This lecture covers several string methods in Python that manipulate the case of text. It begins by introducing the concept of case conversion as part of string class methods and explains the immutability of strings. The instructor explains and demonstrates the following string methods: `capitalize()`, `upper()`, `lower()`, `title()`, `swapcase()`, and `casefold()`. Each method's behavior on example strings is shown using Jupyter Notebook, with notes on how the original string remains unchanged while new strings are returned. Special emphasis is placed on `casefold()` being a more comprehensive lowercase conversion that works well with Unicode and multiple languages. The lecture explains practical use cases where ignoring case differences is important (such as string equality testing) and encourages practicing these methods.

Section 2: Key Concepts and Explanations

- String Case Conversion Methods:

- **`capitalize()`**: Converts only the first character of the string to uppercase and the rest to lowercase.
- **`upper()`**: Converts all characters in the string to uppercase.
- **`lower()`**: Converts all characters in the string to lowercase.
- **`title()`**: Converts the first character of each word in the string to uppercase and the rest in each word to lowercase.
- **`swapcase()`**: Inverts the case of each character; uppercase becomes lowercase, and lowercase becomes uppercase.
- **`casefold()`**: Like `lower()`, it converts all characters to lowercase but more aggressively and supports Unicode, making it ideal for case-insensitive comparisons in diverse languages.

- String Immutability:

- Strings in Python are immutable, meaning string methods do not modify the original string, but instead return a new string reflecting the changes.

- Practical Notes:

- The `casefold()` method is preferred over `lower()` when working with Unicode or internationalized applications.

- For case-insensitive string comparison, converting both strings to the same case with `casefold()` helps ensure accurate equality checks regardless of case differences.

- Relationship to Text Processing Applications:

- These string methods provide functionalities similar to case conversion features found in word processors and text editors.

Section 3: Example Code and Use Cases

Example showing `capitalize()`

```
s1 = 'hello dear'
s2 = s1.capitalize()
print(s2) # Output: Hello dear
```

Explanation: The first letter is capitalized; all others are lowercase.

Example with `capitalize()` when inner letters are uppercase

```
s1 = 'hello Dear'
s2 = s1.capitalize()
print(s2) # Output: Hello dear
```

Explanation: Only the first letter is capitalized; rest are forced lowercase.

Example showing upper()

```
s1 = 'hello dear'  
s2 = s1.upper()  
print(s2) # Output: HELLO DEAR
```

Explanation: All letters converted to uppercase.

Example showing lower()

```
s1 = 'HELLO DEAR'  
s2 = s1.lower()  
print(s2) # Output: hello dear
```

Explanation: All letters converted to lowercase.

Example showing title()

```
s1 = 'hello how are you'  
s2 = s1.title()  
print(s2) # Output: Hello How Are You
```

Explanation: First letter of each word is capitalized; rest are lowercase.

Example showing swapcase()

```
s1 = 'Hello How Are You'  
s2 = s1.swapcase()  
print(s2) # Output: hELLO hOW aRE yOU
```

Explanation: Uppercase letters become lowercase and vice versa.

Example showing casefold()

```
s1 = 'HELLO how are YOU'  
s2 = s1.casefold()  
print(s2) # Output: hello how are you
```

Explanation: Converts string to lowercase suitable for Unicode strings and case-insensitive comparisons.

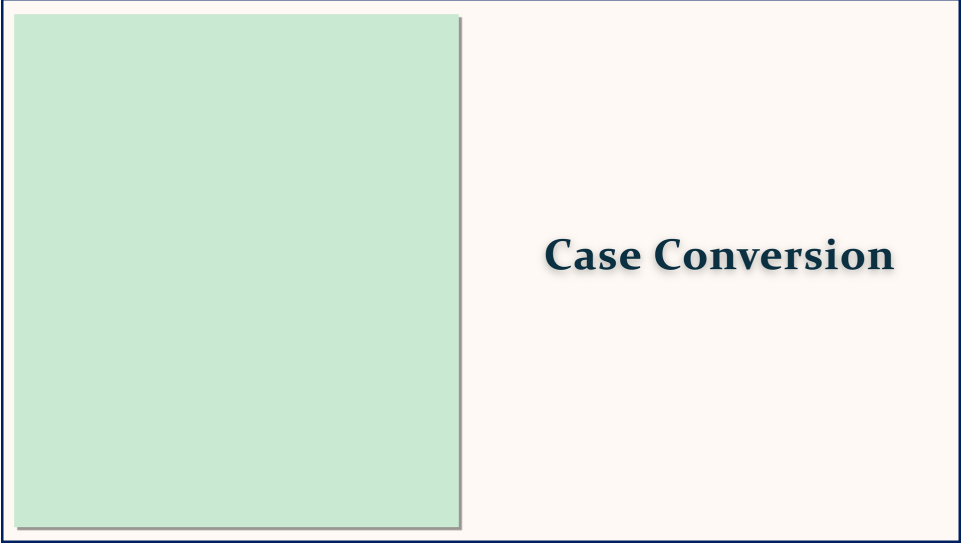
Section 4: Key Takeaways

- Python's string class provides built-in methods to convert string case in various ways: capitalize, upper, lower, title, swapcase, and casefold.
- These methods produce new strings without modifying the original since strings are immutable.
- capitalize() changes only the first character to uppercase; title() changes the first character of every word to uppercase.
- swapcase() toggles the case of each character.
- casefold() is an enhanced version of lower(), useful for Unicode-aware and case-insensitive string comparisons.
- When comparing strings ignoring case, using casefold() on both strings is the most reliable approach.
- These methods are essential tools for text data cleaning, formatting, and normalization in Python programming.

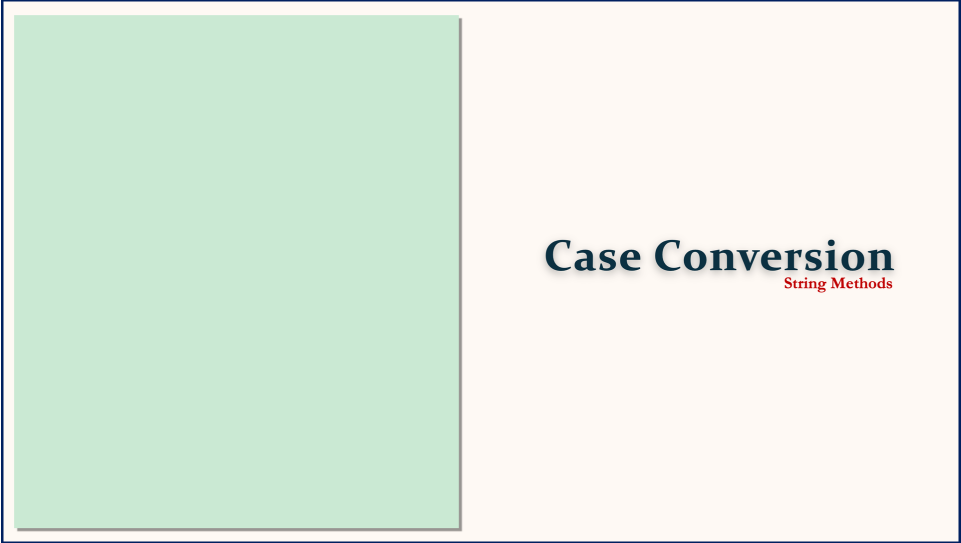
Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





Case Conversion



Case Conversion

String Methods

Case Conversion

String Methods

Case Conversion

String Methods

capitalize()

Case Conversion

String Methods

capitalize()

S1 h e l l o d e a r

Case Conversion

String Methods

capitalize()

upper()

S1 h e l l o d e a r

Case Conversion

String Methods

capitalize()

upper()

S1

H	E	L	L	O		D	E	A	R
---	---	---	---	---	--	---	---	---	---

Case Conversion

String Methods

capitalize()

upper()

lower()

S1

H	E	L	L	O		D	e	A	r
---	---	---	---	---	--	---	---	---	---

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

S1 h e l l o d e a r

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

S1 H E L L O D e A r

Case Conversion

String Methods

capitalize()

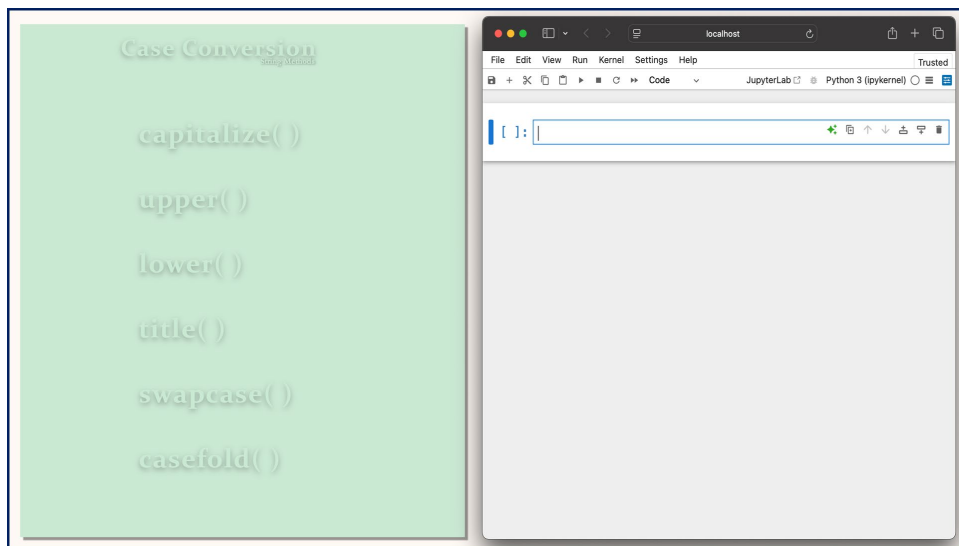
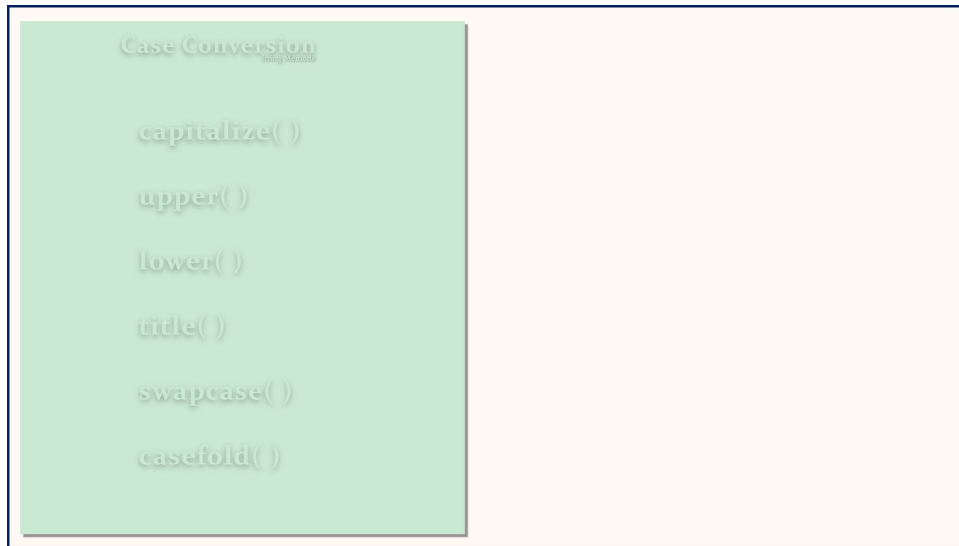
upper()

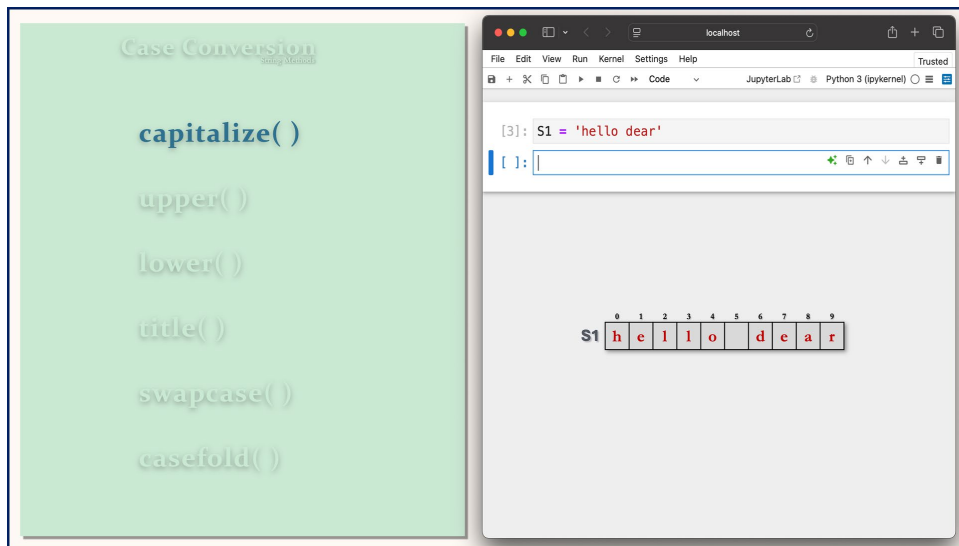
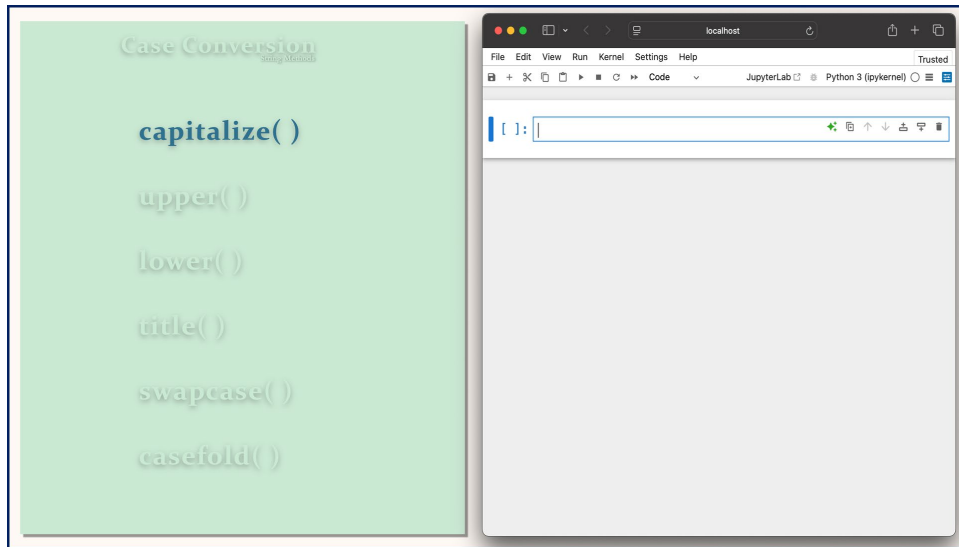
lower()

title()

swapcase()

casefold()





Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (pykernel)

[3]: S1 = 'hello dear'

[]:

S1

0	1	2	3	4	5	6	7	8	9
h	e	l	l	o		d	e	a	r

S2= S1.capitalize()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (pykernel)

[1]: S1 = 'hello dear'

[]:

S1

0	1	2	3	4	5	6	7	8	9
h	e	l	l	o		d	e	a	r

S2= S1.capitalize()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python [conda env:base] * ...

[10...] S1 = 'hello dear'

[10...] S2 = S1.capitalize()

[]:

S1

0 1 2 3 4 5 6 7 8 9

h e l l o d e a r

S2

0 1 2 3 4 5 6 7 8 9

H e l l o d e a r

S2= S1.capitalize()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python [conda env:base] * ...

[10...] S1 = 'hello dear'

[10...] S2 = S1.capitalize()

[10...] print(S2)

Hello dear

[]:

S1

0 1 2 3 4 5 6 7 8 9

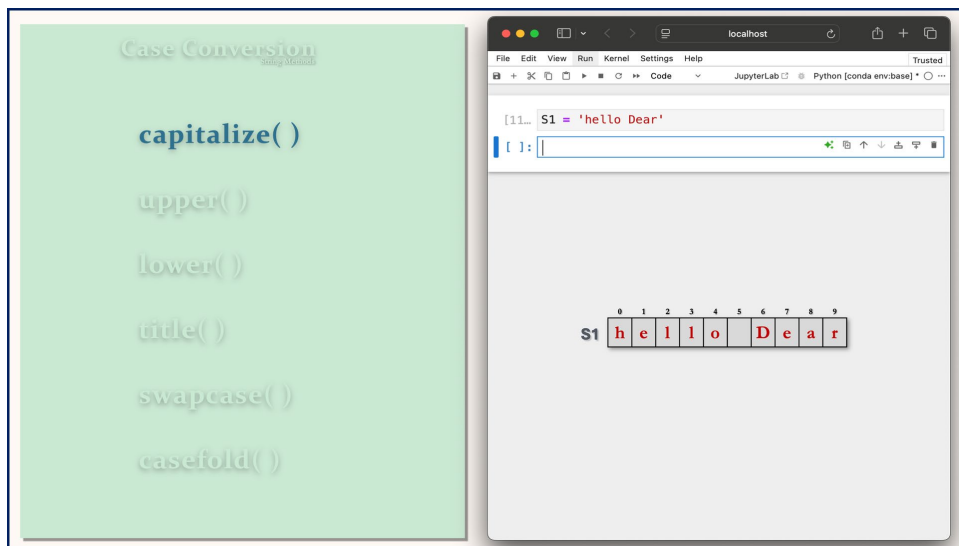
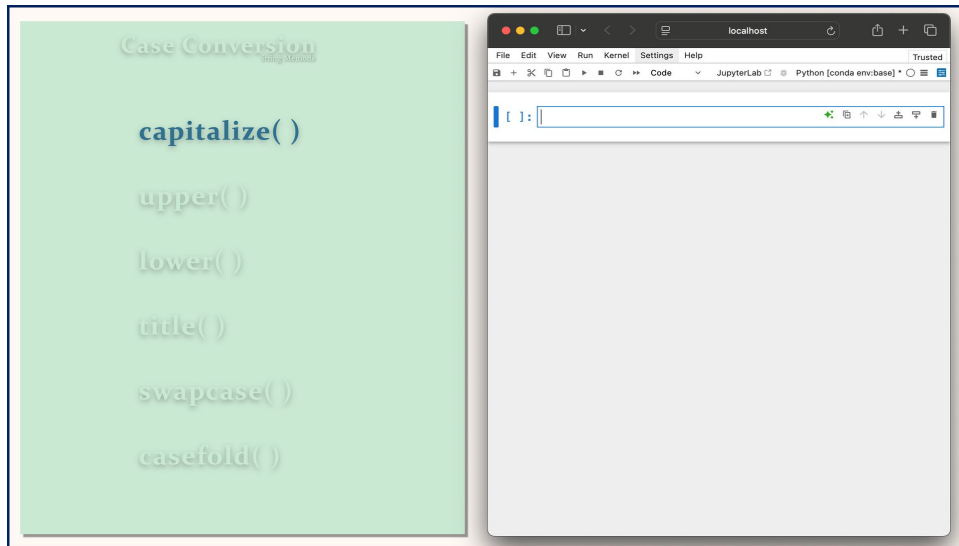
h e l l o d e a r

S2

0 1 2 3 4 5 6 7 8 9

H e l l o d e a r

S2= S1.capitalize()



Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

```
File Edit View Run Kernel Settings Help
JupyterLab Python [conda env:base] *
[11...] S1 = 'hello Dear'
In [ ]:
```

	0	1	2	3	4	5	6	7	8	9
S1	h	e	l	l	o		D	e	a	r

```
S2 = S1.capitalize()
```

Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python [conda env:base]

[11]: S1 = 'hello Dear'

[11]: S2 = S1.capitalize()

[]:

S1

0 1 2 3 4 5 6 7 8 9

h e l l o D e a r

S2

0 1 2 3 4 5 6 7 8 9

H e l l o d e a r

S2 = S1.capitalize()

Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

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JupyterLab Python [conda env:base]

[2]: S1 = 'Hello Dear'

[4]: S2 = S1.capitalize()

[6]: print(S2)

Hello dear

[]:

S1

0 1 2 3 4 5 6 7 8 9

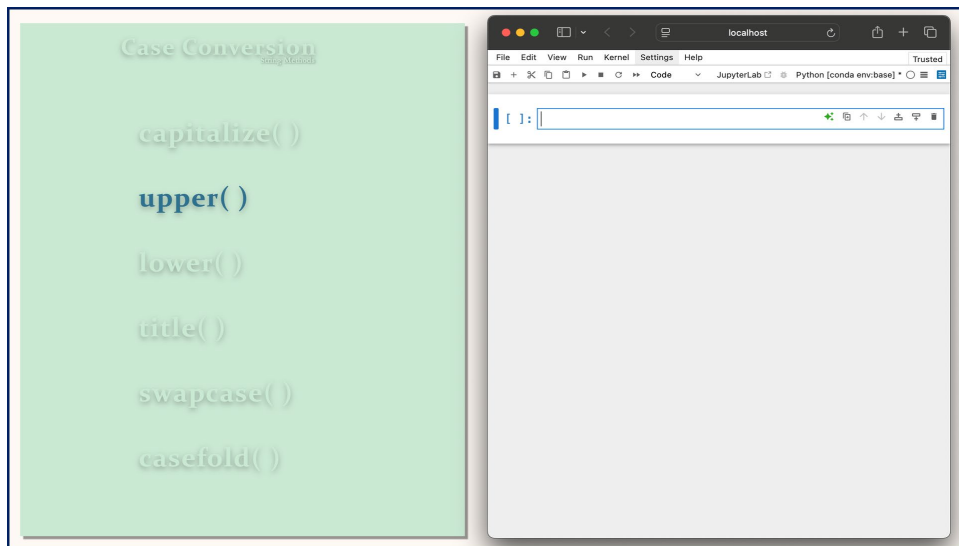
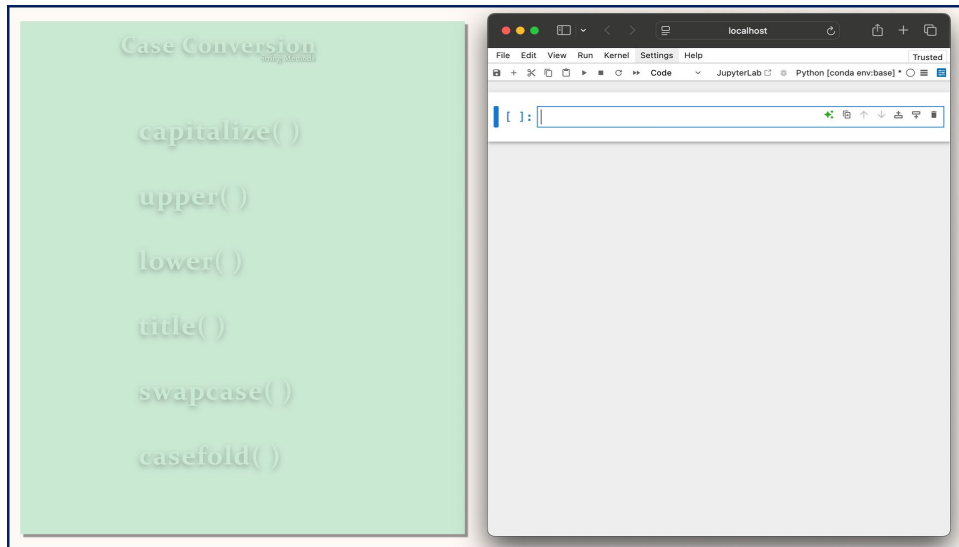
h e l l o D e a r

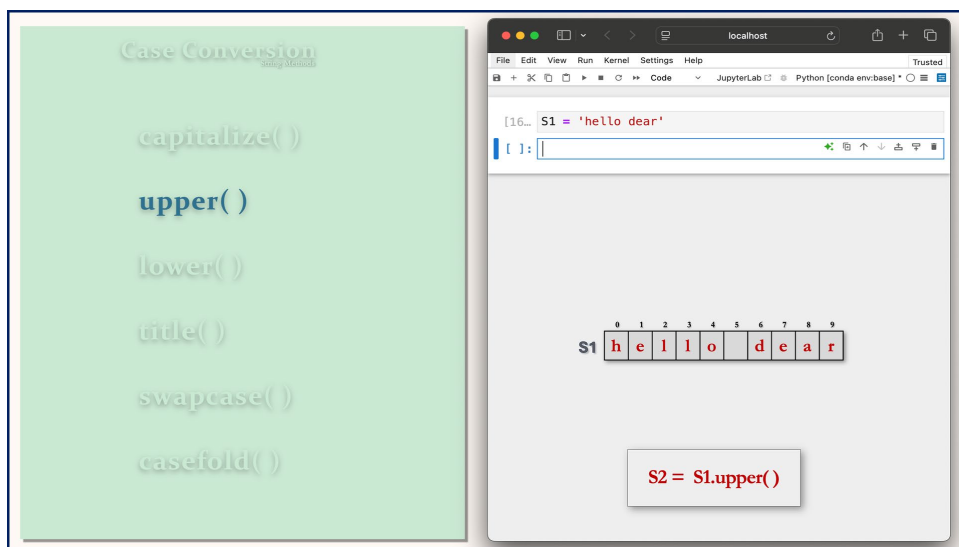
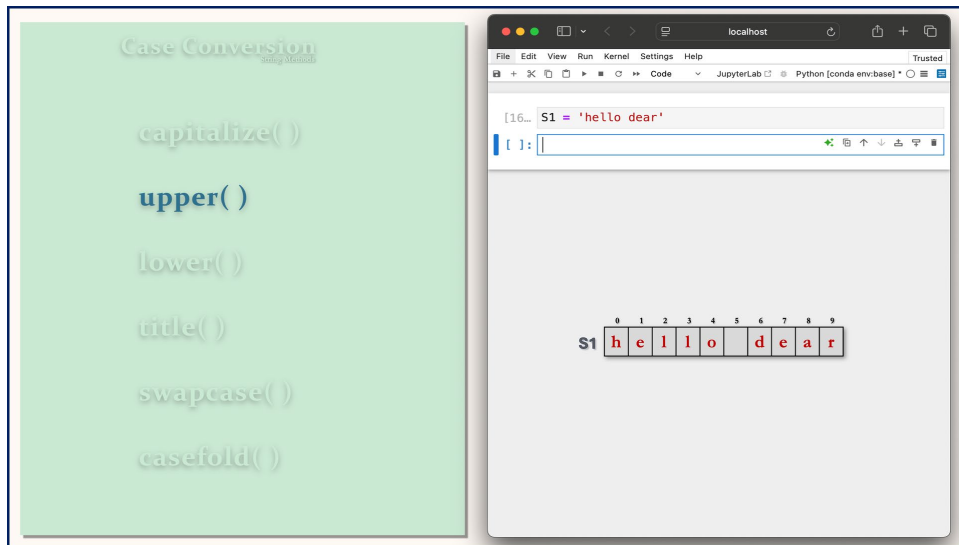
S2

0 1 2 3 4 5 6 7 8 9

H e l l o d e a r

S2 = S1.capitalize()





Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

```
S1 = 'hello dear'
```

```
[16... S1 = 'hello dear'
```

```
[17... S2 = S1.upper()
```

0	1	2	3	4	5	6	7	8	9	
S1	h	e	l	l	o		d	e	a	r

S2 = S1.upper()

Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

```
S1 = 'hello dear'
```

```
[16... S1 = 'hello dear'
```

```
[18... S2 = S1.upper()
```

```
[19... ]
```

0	1	2	3	4	5	6	7	8	9	
S1	h	e	l	l	o		d	e	a	r

0	1	2	3	4	5	6	7	8	9	
S2	H	E	L	L	O		D	E	A	R

S2 = S1.upper()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

+ - X Z [] > < > > > Code JupyterLab Python [conda env:base] * []

[16... S1 = 'hello dear'

[18... S2 = S1.upper()

[20... print(S2)

HELLO DEAR

[]:

S1 h e l l o d e a r

S2 H E L L O D E A R

S2 = S1.upper()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

+ - X Z [] > < > > > Code JupyterLab Python [conda env:base] * []

[]:

Case Conversion

String Methods

`capitalize( )`

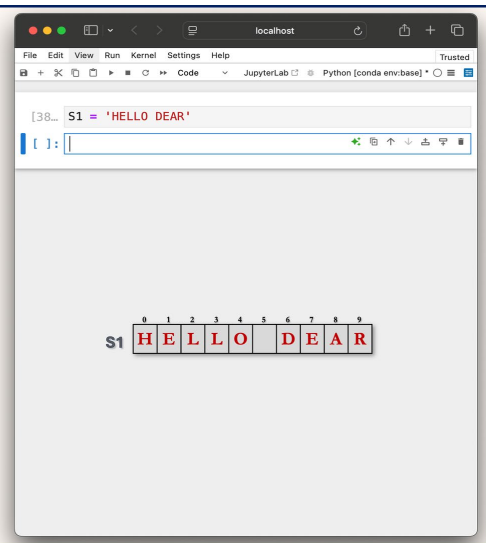
`upper( )`

`lower( )`

`title( )`

`swapcase( )`

`casefold( )`



```
[38...] s1 = 'HELLO DEAR'
```

```
In [ ]:
```

Output:

s1

H	E	L	L	O		D	E	A	R
---	---	---	---	---	--	---	---	---	---

Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base]

[38... S1 = 'HELLO DEAR'

[]:

S1

0	1	2	3	4	5	6	7	8	9
H	E	L	L	O		D	E	A	R

S2 = S1.lower()

Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base]

[38... S1 = 'HELLO DEAR'

[]: S2 = S1.lower()

S1

0	1	2	3	4	5	6	7	8	9
H	E	L	L	O		D	E	A	R

S2 = S1.lower()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

[38... S1 = 'HELLO DEAR'

[40... S2 = S1.lower()

[]:

S1 H E L L O D E A R

S2 h e l l o d e a r

S2 = S1.lower()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

[38... S1 = 'HELLO DEAR'

[40... S2 = S1.lower()

[42... print(S2)

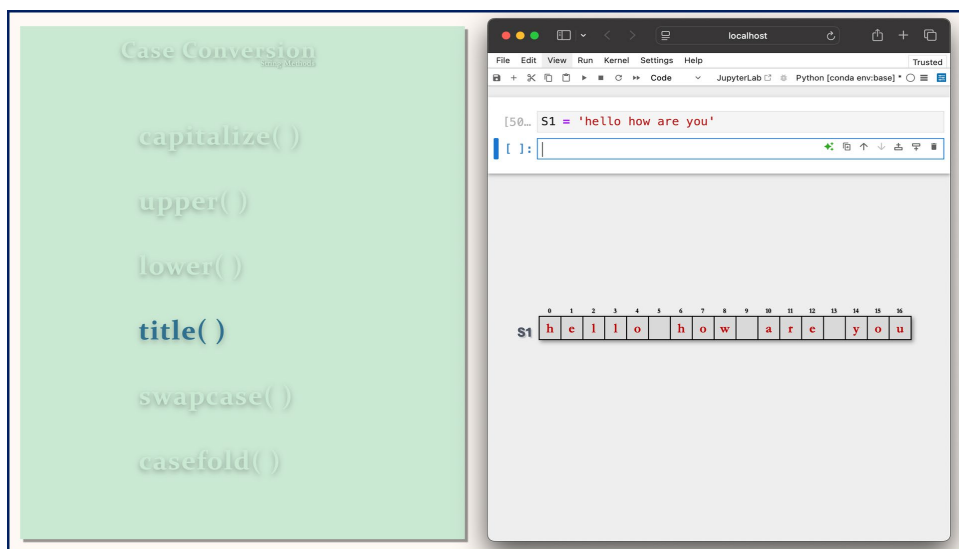
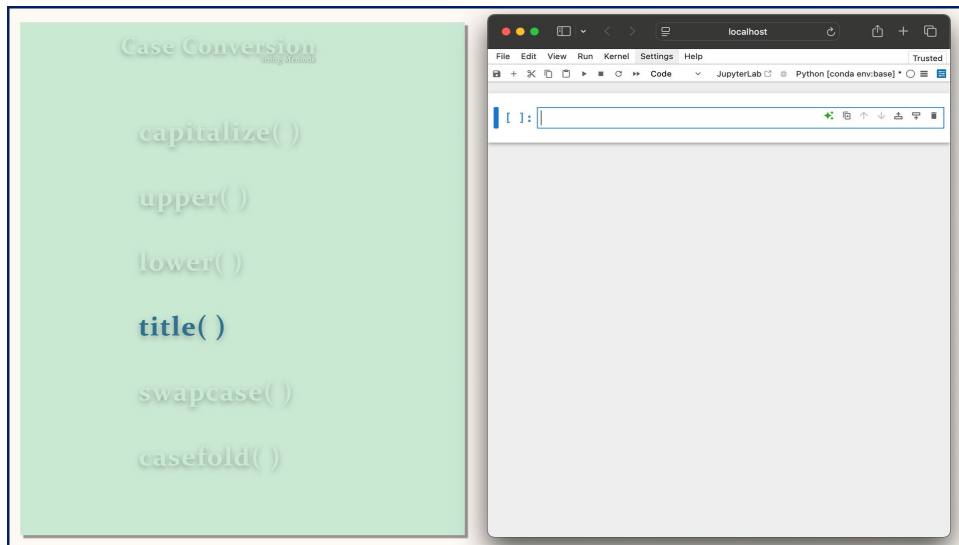
hello dear

[]:

S1 H E L L O D E A R

S2 h e l l o d e a r

S2 = S1.lower()



Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Python [conda env:base] *

[50... S1 = 'hello how are you'

[]:

S1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
h	e	l	l	o		h	o	w		a	r	e		y	o	u

S2 = S1.title()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Python [conda env:base] *

[50... S1 = 'hello how are you'

[]: S2 = S1.title()

S1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
h	e	l	l	o		h	o	w		a	r	e		y	o	u

S2 = S1.title()

Case Conversion
String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

[50...] S1 = 'hello how are you'

[52...] S2 = S1.title()

[]:

S1

S2

S2 = S1.title()

Case Conversion
String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

[50...] S1 = 'hello how are you'

[52...] S2 = S1.title()

[54...] print(S2)

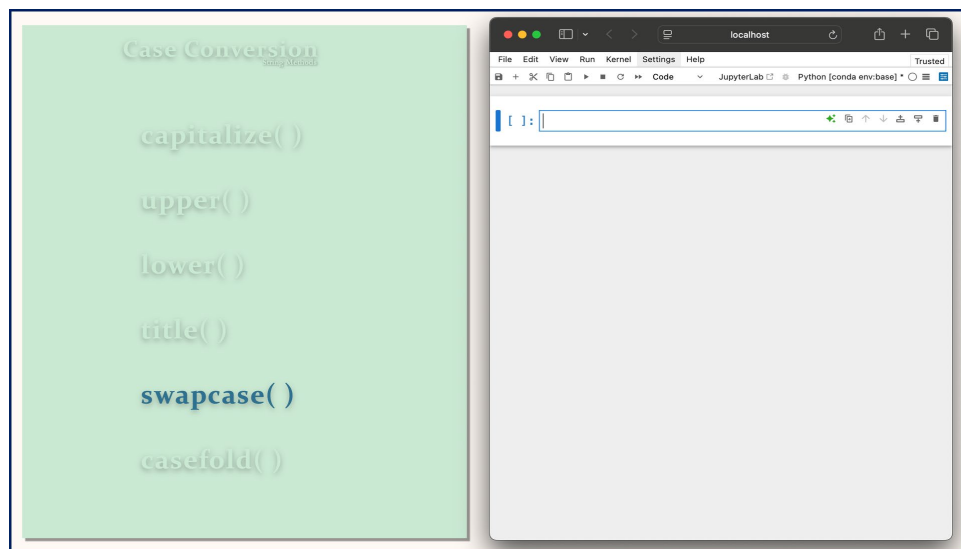
Hello How Are You

[]:

S1

S2

S2 = S1.title()



Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code Python [conda env:base] *

[44... S1 = 'Hello How Are You'

[]:

S1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
H	e	l	l	o		H	o	w		A	r	e		Y	o	u

Case Conversion

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code Python [conda env:base] *

[44... S1 = 'Hello How Are You'

[]:

S1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
H	e	l	l	o		H	o	w		A	r	e		Y	o	u

S2 = S1.swapcase()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base]

[44... S1 = 'Hello How Are You'

[]: S2 = S1.swapcase()

S1 H e l l o H o w A r e Y o u

S2 = S1.swapcase()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base]

[44... S1 = 'Hello How Are You'

[46... S2 = S1.swapcase()

[]:

S1 H e l l o H o w A r e Y o u

S2 h E L L O h O W a R E y O U

S2 = S1.swapcase()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base]

[44...] S1 = 'Hello How Are You'

[46...] S2 = S1.swapcase()

[48...] print(S2)

hELLO hOW aRE yOU

[]:

S1 H e l l o H o w A r e Y o u

S2 h E L L O h O W a R E y O U

S2 = S1.swapcase()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

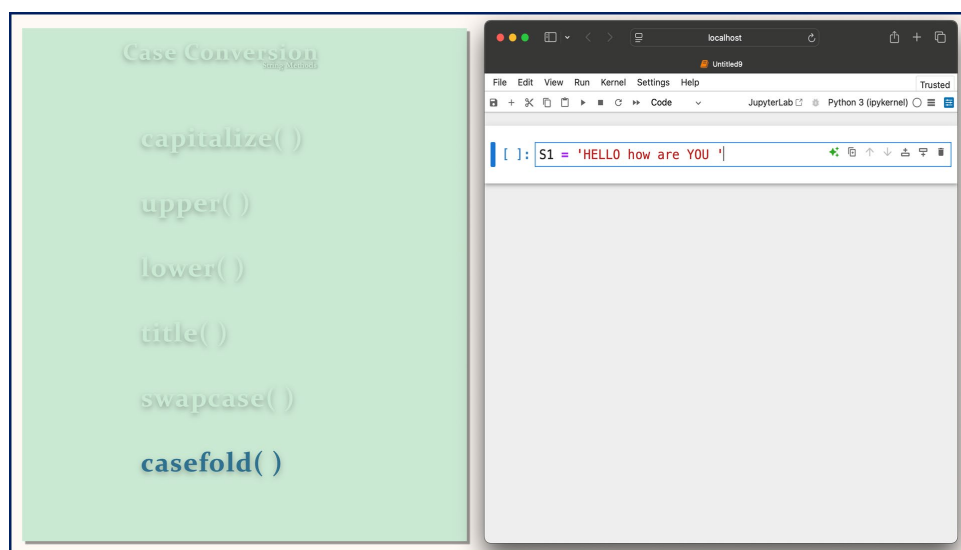
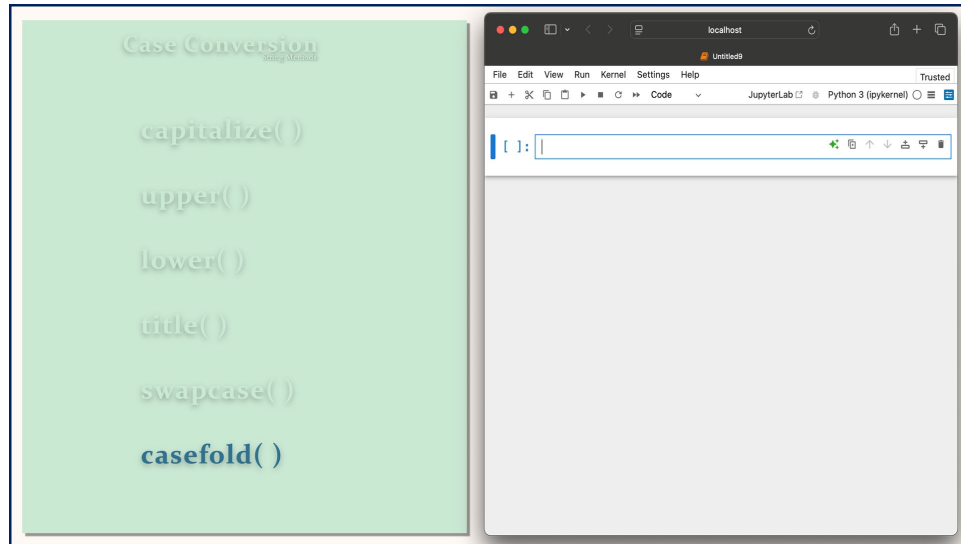
casefold()

localhost

File Edit View Run Kernel Settings Help Untrusted

Code JupyterLab Python 3 (pykernel)

[]:



Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

Untitled9

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (pykernel)

[1]: S1 = 'HELLO how are YOU'

[]:

S1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
H	E	L	L	O		h	o	w		a	r	e		Y	O	U

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

Untitled9

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (pykernel)

[1]: S1 = 'HELLO how are YOU'

[]:

S1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
H	E	L	L	O		h	o	w		a	r	e		Y	O	U

S2 = S1.casefold()

Case Conversion

String Methods

`capitalize( )`

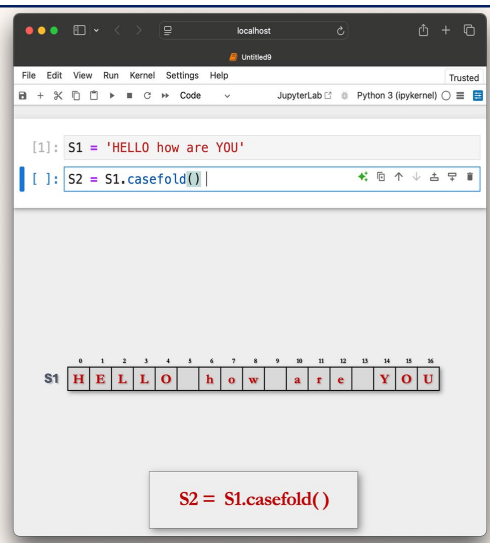
`upper( )`

`lower( )`

`title( )`

`swapcase( )`

`casefold( )`



The screenshot shows a JupyterLab window with the following content:

```
[1]: S1 = 'HELLO how are YOU'
```

```
[ ]: S2 = S1.casefold()
```

Below the code, a string visualization for `S1` is shown:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
S1	H	E	L	L	O		h	o	w		a	r	e		Y	O	U

At the bottom, a box contains the code:

```
S2 = S1.casefold()
```

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

Untitled9

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (pykernel)

[1]: S1 = 'HELLO how are YOU'

[3]: S2 = S1.casefold()

[]:

S1 H E L L O h o w a r e Y O U

S2 h e l l o h o w a r e y o u

S2 = S1.casefold()

Case Conversion

String Methods

capitalize()

upper()

lower()

title()

swapcase()

casefold()

localhost

Untitled9

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (pykernel)

[1]: S1 = 'HELLO how are YOU'

[3]: S2 = S1.casefold()

[5]: print(S2)

hello how are you

[]:

S1 H E L L O h o w a r e Y O U

S2 h e l l o h o w a r e y o u

S2 = S1.casefold()